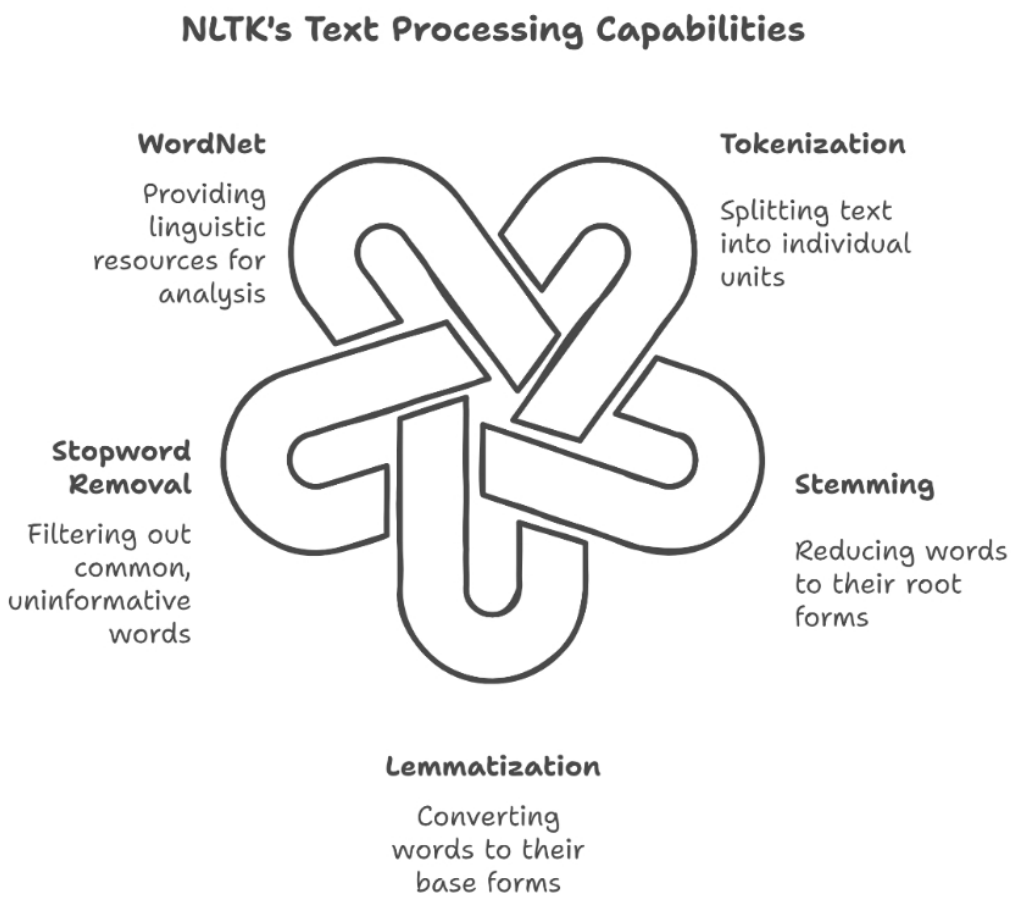


Example: Basic Text Preprocessing with NLTK

```
import nltk
from nltk.tokenize import word_tokenize, sent_tokenize from nltk.corpus import stopwords from nltk.stem import PorterStemmer from nltk.stem import WordNetLemmatizer # Download necessary NLTK data files nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')

text = "Natural Language Processing enables machines to understand human language."

# Tokenization
word_tokens = word_tokenize(text) sent_tokens = sent_tokenize(text) print("Word Tokens:", word_tokens) print("Sentence Tokens:", sent_tokens) # Remove stopwords
stop_words = set(stopwords.words('english')) filtered_words = [word for word in word_tokens if word.lower() not in stop_words]
print("Filtered Words:", filtered_words) # Stemming
ps = PorterStemmer()
stemmed_words = [ps.stem(word) for word in filtered_words]
print("Stemmed Words:", stemmed_words) # Lemmatization
```